

the present. Coupon rates are applied to determine coupon amounts, and the current real value of the accrued principal will be returned as the face value at the maturity date. The nominal ask price combines the ask price with the accrued principal (multiply together and divide by one hundred) to let a purchaser know how much must be paid today in order to purchase the TIPS on the wholesale market. This column is relevant for determining the cost of building the TIPS ladder, but it is a nominal number that is not connected to the yield. The real yield to maturity is provided in inflation-adjusted terms, linking the ask price and accrued principal with the coupon payments and maturity to find the inflation-adjusted rate of return for holding the TIPS.

As mentioned, to construct the ladder, we have to start from the end date. Starting at 2046, we need to buy enough shares of TIPS to give us \$40,000 of inflation-adjusted income that year. This involves buying the TIPS maturing in 2046, which has a coupon of 1 percent, an asking price of \$101.47, a yield of 0.94 percent, and accrued principal of \$1,020. In real terms based on today's accrued principal, and with a simplification that the full coupon payment is made once per year instead of twice, on February 15, 2046, this bond will pay $1020 \times (1 + 0.01) = \$1,030.20$ in real interest and principal. We want an income of \$40,000, so we need to buy $40000 / 1030.20 = 38.8274$ shares. The nominal price we pay for a share of this TIPS today is \$1,034.98. Given the wholesale nominal asking price, these shares cost us $38.8274 \times 1034.98 = \$40,186$ today. Paying \$40,186 today entitles us to \$40,000 of real income in 2046, in addition to \$396.04, which is $0.01 \times 1020 \times 38.83$ shares of real coupon payments for each of the years before then. We know the real value based on today's accrued principal. The nominal amount you actually receive in 2046 will be larger to the extent that we experience inflation over the next thirty years.

Now we move to 2045. We want \$40,000 of real income for that year, too. The trick is that we have to account for the fact that the 2046 maturing TIPS we just purchased is going to give us coupon payments of \$396.04 of income for that year as well. We can subtract that from what we need to purchase. We need an additional \$39,603.96 of real income in 2045. To review the process, the 2045 TIPS has a coupon rate of 0.75 percent and